C++ Workbook

Exercises For Beginners

With Solutions

C++
Workbook: Coding Exercises For Beginners

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1. Write a C++ program to print Hello World.

```
#include <iostream>

int main() {
    std::cout << "Hello World!";
    return 0;
}
```

Output

Hello World

2.Write a C++ program to print number entered by user.

```
#include <iostream>
using namespace std;

int main() {
    int number;

    cout << "Enter an integer: ";
    cin >> number;

    cout << "You entered " << number;
    return 0;
}
```

Output

Enter an integer: 5

You entered: 5

3.Write a C++ program to add two numbers.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int firstNumber, secondNumber, sumOfTwoNumbers;

    cout << "Enter two integers: ";
    cin >> firstNumber >> secondNumber;
    sumOfTwoNumbers = firstNumber + secondNumber;
    cout << firstNumber << " + " <<  secondNumber << " = " << sumOfTwoNumbers;

    return 0;
}
```

Output

Enter two integers: 5
7
5 + 7 = 12

4. Write a C++ program to find quotient and remainder.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int divisor, dividend, quotient, remainder;

    cout << "Enter dividend: ";
    cin >> dividend;

    cout << "Enter divisor: ";
    cin >> divisor;

    quotient = dividend / divisor;
    remainder = dividend % divisor;

    cout << "Quotient = " << quotient << endl;
    cout << "Remainder = " << remainder;

    return 0;
}
```

Output

Enter dividend: 12
Enter divisor: 4

Quotient = 3
Remainder = 0

5. Write a C++ program to find size of int, float, double and char.

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "Size of char: " << sizeof(char) << " byte" << endl;
    cout << "Size of int: " << sizeof(int) << " bytes" << endl;
    cout << "Size of float: " << sizeof(float) << " bytes" << endl;
    cout << "Size of double: " << sizeof(double) << " bytes" << endl;

    return 0;
}
```

Output

Size of char: 1 byte

Size of int: 4 bytes

Size of float: 4 bytes

Size of double: 8 bytes

6. Write a C++ program to swap two numbers using a temporary variable.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a = 5, b = 10, temp;

    cout << "Before swapping." << endl;
    cout << "a = " << a << ", b = " << b << endl;

    temp = a;
    a = b;
    b = temp;

    cout << "\nAfter swapping." << endl;
    cout << "a = " << a << ", b = " << b << endl;

    return 0;
}
```

Output

Before swapping.

a = 5, b = 10

After swapping.

a = 10, b = 5

7. Write a C++ program to swap two numbers without using a temporary variable.

```cpp
#include <iostream>
using namespace std;

int main()
{

    int a = 5, b = 10;

    cout << "Before swapping." << endl;
    cout << "a = " << a << ", b = " << b << endl;

    a = a + b;
    b = a - b;
    a = a - b;

    cout << "\nAfter swapping." << endl;
    cout << "a = " << a << ", b = " << b << endl;

    return 0;
}
```

Output

Before swapping.

a = 5, b = 10

After swapping.

a = 10, b = 5

8. Write a C++ program to check whether a character is vowel or consonant.

```cpp
#include <iostream>
using namespace std;

int main() {
    char c;
    bool isLowercaseVowel, isUppercaseVowel;

    cout << "Enter an alphabet: ";
    cin >> c;

    isLowercaseVowel = (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u');

    isUppercaseVowel = (c == 'A' || c == 'E' || c == 'I' || c == 'O' || c == 'U');

    if (!isalpha(c))
      printf("Error! Non-alphabetic character.");
    else if (isLowercaseVowel || isUppercaseVowel)
        cout << c << " is a vowel.";
    else
        cout << c << " is a consonant.";

    return 0;
}
```

Output

Enter an alphabet: E

E is a vowel

9. Write a C++ program to find largest number among three numbers.

```cpp
#include <iostream>
using namespace std;

int main() {
    float n1, n2, n3;

    cout << "Enter three numbers: ";
    cin >> n1 >> n2 >> n3;

    if(n1 >= n2 && n1 >= n3)
        cout << "Largest number: " << n1;

    if(n2 >= n1 && n2 >= n3)
        cout << "Largest number: " << n2;

    if(n3 >= n1 && n3 >= n2)
        cout << "Largest number: " << n3;

    return 0;
}
```

Output

Enter three numbers:

8

7

5

Largest number: 8

10. Write a C++ program to find all roots of a quadratic equation.

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {

    float a, b, c, x1, x2, discriminant, realPart, imaginaryPart;
    cout << "Enter coefficients a, b and c: ";
    cin >> a >> b >> c;
    discriminant = b*b - 4*a*c;

    if (discriminant > 0) {
        x1 = (-b + sqrt(discriminant)) / (2*a);
        x2 = (-b - sqrt(discriminant)) / (2*a);
        cout << "Roots are real and different." << endl;
        cout << "x1 = " << x1 << endl;
        cout << "x2 = " << x2 << endl;
    }

    else if (discriminant == 0) {
        cout << "Roots are real and same." << endl;
        x1 = -b/(2*a);
        cout << "x1 = x2 =" << x1 << endl;
    }

    else {
        realPart = -b/(2*a);
        imaginaryPart =sqrt(-discriminant)/(2*a);
        cout << "Roots are complex and different."  << endl;
        cout << "x1 = " << realPart << "+" << imaginaryPart << "i" << endl;
        cout << "x2 = " << realPart << "-" << imaginaryPart << "i" << endl;
    }

    return 0;
}
```

Output

Enter coefficients a, b and c:

4

5

1

Roots are real and different.

x1 = -0.25

x2 = -1

11. Write a C++ program to find sum of natural numbers.

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;

    cout << "Enter a positive integer: ";
    cin >> n;

    for (int i = 1; i <= n; ++i) {
        sum += i;
    }

    cout << "Sum = " << sum;
    return 0;
}

```

Output

Enter a positive integer: 50

Sum = 1275

12. Write a C++ program to check leap year.

```cpp
#include <iostream>
using namespace std;

int main() {
    int year;

    cout << "Enter a year: ";
    cin >> year;

    if (year % 4 == 0) {
        if (year % 100 == 0) {
            if (year % 400 == 0)
                cout << year << " is a leap year.";
            else
                cout << year << " is not a leap year.";
        }
        else
            cout << year << " is a leap year.";
    }
    else
        cout << year << " is not a leap year.";

    return 0;
}

```

Output

Enter a year: 2014

2014 is not a leap year

13. Write a C++ program to find factorial.

```
#include <iostream>
using namespace std;

int main() {
    int n;
    long double factorial = 1.0;

    cout << "Enter a positive integer: ";
    cin >> n;

    if (n < 0)
        cout << "Error! Factorial of a negative number doesn't exist.";
    else {
        for(int i = 1; i <= n; ++i) {
            factorial *= i;
        }
        cout << "Factorial of " << n << " = " << factorial;
    }

    return 0;
}
```

Output

Enter a positive integer: 12

Factorial of 12 = 479001600

14. Write a C++ program to generate multiplication table.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, range;

    cout << "Enter an integer: ";
    cin >> n;

    cout << "Enter range: ";
    cin >> range;

    for (int i = 1; i <= range; ++i) {
        cout << n << " * " << i << " = " << n * i << endl;
    }

    return 0;
}

```

Output

Enter an integer: 3

Enter range: 8

3 * 1 = 3

3 * 2 = 6

3 * 3 = 9

3 * 4 = 12

3 * 5 = 15

3 * 6 = 18

3 * 7 = 21

3 * 8 = 24

15. Write a C++ program to generate Fibonacci series.

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, t1 = 0, t2 = 1, nextTerm = 0;

    cout << "Enter the number of terms: ";
    cin >> n;

    cout << "Fibonacci Series: ";

    for (int i = 1; i <= n; ++i) {
        if(i == 1) {
            cout << t1 << ", ";
            continue;
        }
        if(i == 2) {
            cout << t2 << ", ";
            continue;
        }
        nextTerm = t1 + t2;
        t1 = t2;
        t2 = nextTerm;

        cout << nextTerm << ", ";
    }
    return 0;
}
```

Output

Enter the number of terms: 9

Fibonacci Series: 0,1,1,2,3,5,8,13,21

16. Write a C++ program to find GCD/HCF.

```cpp
#include <iostream>
using namespace std;

int main() {
  int n1, n2;

  cout << "Enter two numbers: ";
  cin >> n1 >> n2;

  while(n1 != n2) {
    if(n1 > n2)
      n1 -= n2;
    else
      n2 -= n1;
  }

  cout << "HCF = " << n1;

  return 0;
}
```

Output

Enter two numbers:

16

76

HCF = 4

17. Write a C++ program to find LCM.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n1, n2, max;

    cout << "Enter two numbers: ";
    cin >> n1 >> n2;

    max = (n1 > n2) ? n1 : n2;

    do
    {
        if (max % n1 == 0 && max % n2 == 0)
        {
            cout << "LCM = " << max;
            break;
        }
        else
            ++max;
    } while (true);

    return 0;
}
```

Output

```
Enter two numbers:
12
18
LCM = 36
```

18. Write a C++ program to reverse a number.

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, reversedNumber = 0, remainder;

    cout << "Enter an integer: ";
    cin >> n;

    while(n != 0) {
        remainder = n%10;
        reversedNumber = reversedNumber*10 + remainder;
        n /= 10;
    }

    cout << "Reversed Number = " << reversedNumber;

    return 0;
}
```

Output

Enter an integer: 1234

Reversed Number = 4321

19. Write a C++ program to calculate power of a number.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int exponent;
    float base, result = 1;

    cout << "Enter base and exponent respectively:  ";
    cin >> base >> exponent;

    cout << base << "^" << exponent << " = ";

    while (exponent != 0) {
        result *= base;
        --exponent;
    }

    cout << result;

    return 0;
}
```

Output

Enter base and exponent respectively:

4

2

4^2 = 16

20. Write a C++ program to multiply two numbers.

```
#include <iostream>
using namespace std;

int main() {
  double num1, num2, product;
  cout << "Enter two numbers: ";

  cin >> num1 >> num2;

  product = num1 * num2;

  cout << "Product = " << product;

  return 0;
}
```

Output

Enter two numbers:

4

5

Product = 20

21. Write a C++ program to check whether a number is palindrome or not.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, num, digit, rev = 0;

    cout << "Enter a positive number: ";
    cin >> num;

    n = num;

    do
    {
        digit = num % 10;
        rev = (rev * 10) + digit;
        num = num / 10;
    } while (num != 0);

    cout << " The reverse of the number is: " << rev << endl;

    if (n == rev)
        cout << " The number is a palindrome.";
    else
        cout << " The number is not a palindrome.";

    return 0;
}
```

Output

Enter a positive number: 12321

The reverse of the number is: 12321

The number is a palindrome.

22. Write a C++ program to check whether a number is prime or not.

```cpp
#include <iostream>
using namespace std;

int main() {
    int i, n;
    bool isPrime = true;

    cout << "Enter a positive integer: ";
    cin >> n;

    if (n == 0 || n == 1) {
        isPrime = false;
    }
    else {
        for (i = 2; i <= n / 2; ++i) {
            if (n % i == 0) {
                isPrime = false;
                break;
            }
        }
    }
    if (isPrime)
        cout << n << " is a prime number";
    else
        cout << n << " is not a prime number";

    return 0;
}
```

Output

Enter a positive integer: 29

29 is a prime number

23. Write a C++ program to check Armstrong number.

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, originalNum, remainder, result = 0;
    cout << "Enter a three-digit integer: ";
    cin >> num;
    originalNum = num;

    while (originalNum != 0) {
        remainder = originalNum % 10;

        result += remainder * remainder * remainder;

        originalNum /= 10;
    }

    if (result == num)
        cout << num << " is an Armstrong number.";
    else
        cout << num << " is not an Armstrong number.";

    return 0;
}
```

Output

Enter a three-digit integer:

371

371 is an Armstrong number

24. Write a C++ program to display factors of a number.

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, i;

    cout << "Enter a positive integer: ";
    cin >> n;

    cout << "Factors of " << n << " are: ";
    for(i = 1; i <= n; ++i) {
        if(n % i == 0)
            cout << i << " ";
    }

    return 0;
}

```

Output

Enter a positive integer:

60

Factors of 60 are:

1 2 3 4 5 6 10 12 15 20 30 60

25. Write a C++ program to make a simple calculator.

```cpp
# include <iostream>
using namespace std;

int main() {

  char op;
  float num1, num2;

  cout << "Enter operator: +, -, *, /: ";
  cin >> op;

  cout << "Enter two operands: ";
  cin >> num1 >> num2;

  switch(op) {

    case '+':
      cout << num1 << " + " << num2 << " = " << num1 + num2;
      break;

    case '-':
      cout << num1 << " - " << num2 << " = " << num1 - num2;
      break;

    case '*':
      cout << num1 << " * " << num2 << " = " << num1 * num2;
      break;

    case '/':
      cout << num1 << " / " << num2 << " = " << num1 / num2;
      break;
    default:
      cout << "Error! operator is not correct";
      break;
  }

  return 0;
}
```

Output

Enter operator: +, -, *, / :

Enter two operands: 5 2

5-2 = 3

26. Write a C++ program to find sum of natural numbers using recursion.

```cpp
#include<iostream>
using namespace std;

int add(int n);

int main() {
    int n;

    cout << "Enter a positive integer: ";
    cin >> n;

    cout << "Sum =  " << add(n);

    return 0;
}

int add(int n) {
    if(n != 0)
        return n + add(n - 1);
    return 0;
}
```

Output

Enter a positive integer: 10

Sum = 55

27. Write a C++ program to find factorial of a number using recursion.

```cpp
#include<iostream>
using namespace std;

int factorial(int n);

int main()
{
    int n;

    cout << "Enter a positive integer: ";
    cin >> n;

    cout << "Factorial of " << n << " = " << factorial(n);

    return 0;
}

int factorial(int n)
{
    if(n > 1)
        return n * factorial(n - 1);
    else
        return 1;
}

```

Output

Enter a positive integer: 6

Factorial of 6 = 720

28. Write a C++ program to find gcd using recursion.

```cpp
#include <iostream>
using namespace std;

int hcf(int n1, int n2);

int main()
{
    int n1, n2;

    cout << "Enter two positive integers: ";
    cin >> n1 >> n2;

    cout << "H.C.F of " << n1 << " & " <<  n2 << " is: " << hcf(n1, n2);

    return 0;
}

int hcf(int n1, int n2)
{
    if (n2 != 0)
        return hcf(n2, n1 % n2);
    else
        return n1;
}

```

Output

Enter two positive integers:

366

60

H.C.F of 366 and 60 is : 6

29. Write a C++ program to find the frequency of characters in a string.

```cpp
#include <iostream>
using namespace std;

int main()
{
    string str = "C++ Programming is awesome";
    char checkCharacter = 'a';
    int count = 0;

    for (int i = 0; i < str.size(); i++)
    {
        if (str[i] ==  checkCharacter)
        {
            ++ count;
        }
    }

    cout << "Number of " << checkCharacter << " = " << count;

    return 0;
}

```

Output

Number of a = 2

30. Write a C++ program to find the length of a string.

```cpp
#include <iostream>
using namespace std;

int main() {
    string str = "C++ Programming";

    cout << "String Length = " << str.size();

    return 0;
}

```

Output

String Length = 15

31. Write a C++ program to concatenate two strings.

```cpp
#include <iostream>
using namespace std;

int main()
{
    string s1, s2, result;

    cout << "Enter string s1: ";
    getline (cin, s1);

    cout << "Enter string s2: ";
    getline (cin, s2);

    result = s1 + s2;

    cout << "Resultant String = "<< result;

    return 0;
}
```

Output

Enter string s1: c++ programming

Enter string s2: is awesome

Resultant String = : c++ programming is awesome

32. Write a C++ program to copy strings.

```cpp
#include <iostream>
using namespace std;

int main()
{
    string s1, s2;

    cout << "Enter string s1: ";
    getline (cin, s1);

    s2 = s1;

    cout << "s1 = "<< s1 << endl;
    cout << "s2 = "<< s2;

    return 0;
}

```

Output

Enter string s1: c++

s1 = c++

s2 = c++

33. Write a C++ program to convert binary number to decimal.

```cpp
#include <iostream>
#include <cmath>

using namespace std;

int convert(long long);

int main() {
  long long n;
  cout << "Enter a binary number: ";
  cin >> n;
  cout << n << " in binary = " << convert(n) << " in decimal";
  return 0;
}

int convert(long long n) {
  int dec = 0, i = 0, rem;

  while (n!=0) {
    rem = n % 10;
    n /= 10;
    dec += rem * pow(2, i);
    ++i;
  }

  return dec;
}
```

Output

Enter a binary number: 1101

1101 in binary = 13 in decimal

34. Write a C++ program to convert octal number to decimal.

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int octalToDecimal(int octalNumber);

int main()
{
   int octalNumber;
   cout << "Enter an octal number: ";
   cin >> octalNumber;
   cout << octalNumber << " in octal = " << octalToDecimal(octalNumber) << " in decimal";

   return 0;
}
int octalToDecimal(int octalNumber)
{
    int decimalNumber = 0, i = 0, rem;
    while (octalNumber != 0)
    {
        rem = octalNumber % 10;
        octalNumber /= 10;
        decimalNumber += rem * pow(8, i);
        ++i;
    }
    return decimalNumber;
}
}
```

Output

Enter an octal number: 2341

2341 in octal = 1249 in decimal

35. Write a C++ program to convert binary number to octal.

```cpp
#include <iostream>
#include <cmath>

using namespace std;

int convertBinarytoOctal(long long);
int main()
{
    long long binaryNumber;
    cout << "Enter a binary number: ";
    cin >> binaryNumber;
    cout << binaryNumber << " in binary = " << convertBinarytoOctal(binaryNumber) << " in octal ";
    return 0;
}

int convertBinarytoOctal(long long binaryNumber)
{
    int octalNumber = 0, decimalNumber = 0, i = 0;
    while(binaryNumber != 0)
    {
        decimalNumber += (binaryNumber%10) * pow(2,i);
        ++i;
        binaryNumber/=10;
    }
    i = 1;
    while (decimalNumber != 0)
    {
        octalNumber += (decimalNumber % 8) * i;
        decimalNumber /= 8;
        i *= 10;
    }
    return octalNumber;
}
```

Output

Enter a binary number: 10001

10001 in binary = 21 in octal